

Enhancing the Interpretability of the Cell-Type Deconvolution Algorithm DISSECT Using SHAP Values

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Abstract—Deep learning models for bulk RNA-seq deconvolution have demonstrated strong predictive performance but remain difficult to interpret biologically. We apply SHAP (SHapley Additive exPlanations) to DISSECT, a deep learning deconvolution framework, to identify gene features driving predicted cell-type proportions across nine immune cell types in healthy and systemic lupus erythematosus (SLE) pseudo-bulk samples. Using a systematic background-variation strategy, we decompose SHAP attributions into three biologically interpretable components: raw cell-type marker signal, domain shift between simulated training data and real tissue, and condition-specific changes in SLE relative to healthy. We show that uninformative zero backgrounds recover canonical immune markers more reliably than noise backgrounds, while simulated pseudo-bulk backgrounds reveal a consistent domain shift shared across healthy and SLE conditions (55 % overlap in top attributed genes). A real healthy background applied to SLE samples isolates disease-specific signals, including monocyte inflammatory pathways, NK cell chemokine signaling, and pDC-driven interferon responses consistent with known SLE pathophysiology. Across conditions, DISSECT preferentially attributes importance to discriminative genes that resolve ambiguity between similar cell types rather than canonical identity markers, reflecting its multi-class training objective. These findings demonstrate that SHAP with principled background selection is a practical and biologically informative framework for interpreting neural network deconvolution models, and establish a generalizable pipeline applicable to other deep learning transcriptomic tools

Index Terms—Explainable Deep Learning, DISSECT, Cell De-convolution, SHAP values

I. Introduction

RNA sequencing (RNA-seq) has revolutionized transcriptomics by enabling transcriptome-wide quantification of gene expression in biological samples. However, bulk RNA-seq measurements represent averaged expression signals across heterogeneous mixtures of cells, obscuring the underlying cellular composition that is often critical to understanding disease mechanisms, immune dynamics, and therapeutic response. Traditionally, experimental techniques such as flow cytometry and immunohistochemistry have been used to characterize cellular composition within tissues, but these approaches are often labor-intensive, costly, and may result in substantial sample loss. To address these limitations, computational deconvolution methods have been developed to estimate cell-type proportions directly from bulk transcriptomic data.

Existing deconvolution approaches generally fall into two categories: reference-free and reference-based methods [1], [2]. Reference-free methods infer latent cellular components directly from bulk expression profiles using unsupervised approaches such as matrix factorization, but often produce less biologically interpretable results. In contrast, reference-based methods leverage predefined cell-type expression signatures, increasingly derived from single-cell RNA sequencing (scRNA-seq), to estimate cellular composition with improved accuracy and resolution. The growing availability of scRNA-seq datasets has accelerated the development of numerous reference-based deconvolution frameworks, which broadly fall into three categories: Regression-based, Bayesian, and Deep learning approaches. Regression-based methods model bulk RNA-seq expression as a linear combination of cell-type-specific reference profiles and include widely used methods such as CIBERSORTx [3], Bisque [4], DWLS [5], and MuSiC [6]. Bayesian approaches, including BayesPrism [7] and InstaPrism [8], incorporate scRNA-seq information within probabilistic frameworks to model uncertainty and variability in the deconvolution process. More recently, deep learning methods such as Scaden [9], DigitalDLsorter [10], and DISSECT [11] have emerged, leveraging neural networks to capture complex nonlinear relationships between bulk and single-cell transcriptomic data. Recent benchmarking studies have shown that DISSECT achieves strong performance under matched, mismatched, and mixed scRNA-seq reference settings [2].

DISSECT uses simulated pseudo-bulk mixtures generated from scRNA-seq reference datasets to train neural networks capable of predicting cell-type proportions from bulk transcriptomic profiles. Unlike traditional regression-based methods that primarily assume linear relationships between bulk and reference expression profiles, DISSECT leverages multilayer perceptron (MLP) architectures to model complex nonlinear relationships within high-dimensional transcriptomic data. In addition to estimating cell-type fractions, DISSECT also incorporates a conditional autoencoder framework to infer cell-type-specific gene expression profiles from bulk samples. An overview of the DISSECT framework is shown in Supplementary Figure 1.

A key feature of DISSECT is its use of semi-supervised consistency regularization to improve robustness under

domain shifts between training and target datasets. In practice, scRNA-seq reference datasets may differ from bulk RNA-seq samples due to disease state, experimental platform, batch effects, or biological heterogeneity. To address this challenge, DISSECT introduces a consistency loss that encourages stable predictions across perturbed or mixed samples during training, improving generalization across diverse datasets. Benchmark evaluations across peripheral blood mononuclear cell (PBMC), pancreas, kidney, and brain datasets demonstrated that DISSECT consistently achieved state-of-the-art performance relative to existing regression-based, Bayesian, and deep learning approaches, while also showing improved robustness to reference mismatch and domain shift.

Despite these performance advantages, DISSECT, like many deep learning models, functions largely as a black-box framework. While the model can accurately estimate cell-type proportions, it remains difficult to determine which genes or transcriptomic features most strongly influence its predictions. This lack of interpretability limits biological insight and may reduce confidence in downstream analyses, particularly in biomedical applications where understanding the molecular drivers of predictions is essential. To address this limitation, we investigate the use of SHAP (SHapley Additive exPlanations) [12] values to improve the interpretability of DISSECT by identifying the gene features that contribute most strongly to predicted cell-type proportions.

II. Related Work

Unlike regression-based approaches, where coefficients may directly correspond to gene contributions, neural network models learn complex nonlinear representations that are often difficult to interpret biologically. As a result, recent work has focused increasingly on incorporating explainability methods into transcriptomic deconvolution frameworks.

One recent example is Sweetwater [13], an adaptive autoencoder-based deconvolution framework that integrates interpretability analysis using DeepLIFT [14]. DeepLIFT (Deep Learning Important FeaTures) is a feature attribution method that assigns contribution scores to input features by comparing neuron activations against a reference baseline and propagating activation differences backward through the network. Unlike standard gradient-based attribution methods, DeepLIFT can better handle issues such as gradient saturation and inactive neurons by explicitly measuring changes relative to a reference activation state. In the Sweetwater framework, DeepLIFT scores were used to identify genes contributing most strongly to predicted cell-type proportions, enabling downstream functional enrichment analysis and recovery of biologically meaningful marker genes.

DeepLIFT is also closely related to SHAP, particularly through the RevealCancel propagation rule introduced in the DeepLIFT framework. The RevealCancel rule improves attribution estimation by separately accounting for positive and negative feature contributions and has been

shown to approximate Shapley values more accurately than simpler linear propagation rules. However, while DeepLIFT provides an efficient backpropagation-based attribution strategy for neural networks, SHAP offers a broader game-theoretic framework with stronger theoretical guarantees, including local accuracy, consistency, and fair feature attribution across all possible feature subsets. This is particularly relevant in transcriptomic applications, where correlated gene expression and complex gene-gene interactions can make attribution unstable under simpler feature importance approaches.

Motivated by these advances, our work investigates the use of SHAP values to improve interpretability of DISSECT, a deep learning-based deconvolution framework that has demonstrated strong robustness and predictive accuracy across diverse deconvolution settings. By quantifying the contribution of individual genes to predicted cell-type proportions, SHAP-based explanations may provide greater biological insight into the transcriptomic patterns learned by DISSECT while improving transparency and reliability of model predictions.

III. Methods

A. SHAP for Model Interpretability

SHAP (SHapley Additive exPlanations) is a unified framework for explaining machine learning model predictions, grounded in cooperative game theory [12]. Given a model $f : \mathbb{R}^p \rightarrow \mathbb{R}$ and an input $\mathbf{x} \in \mathbb{R}^p$ with p features, SHAP assigns each feature j an attribution value ϕ_j such that the model output decomposes additively as:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^p \phi_j \quad (1)$$

where $\phi_0 = \mathbb{E}[f(\mathbf{x})]$ is the baseline expected model output over the background distribution. Each attribution ϕ_j is the Shapley value of feature j , defined as its average marginal contribution across all possible subsets $S \subseteq F \setminus \{j\}$ of the remaining features:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x})] \quad (2)$$

where F is the full set of features, $|F| = p$, and $f_S(\mathbf{x})$ denotes the expected model output when only the features in S are observed, with remaining features marginalized out.

Exact computation of Shapley values requires evaluating 2^p feature subsets, which is computationally intractable for genome-scale RNA-seq data ($p \sim 10^4$ genes). For deep learning models, the GradientExplainer in the SHAP library approximates Shapley values using expected gradients [15], which reformulates the integrated gradients method as an expectation over a background dataset $\mathcal{D}$:

$$\phi_j \approx \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}, \alpha \sim U[0,1]} \left[(x_j - x'_j) \cdot \frac{\partial f(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_j} \right] \quad (3)$$

where $\mathbf{x}'$ is a reference sample drawn from the background distribution and α is a random interpolation coefficient sampled uniformly along the path from $\mathbf{x}'$ to the input $\mathbf{x}$. This formulation allows an entire dataset to serve as the background reference rather than a single fixed baseline, and converges to attributions that satisfy the additive decomposition in Eq. 1.

Importantly, expected gradients satisfies a set of interpretability axioms that make it particularly well-suited for deep learning-based genomic models [15]:

- **Implementation invariance:** Two networks that produce identical outputs for all inputs will receive identical attributions. Any gradient-based attribution method satisfies this axiom, ensuring that attributions reflect functional behavior rather than architectural details.
- **Sensitivity (Dummy):** If a model’s output does not depend on a feature at all, that feature receives zero attribution. This holds for expected gradients because the gradient with respect to an irrelevant feature is zero everywhere.
- **Completeness:** Attributions sum to the difference between the model output at the input and the model output averaged over all background baselines:

$$\sum_{j=1}^p \phi_j = f(\mathbf{x}) - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}}[f(\mathbf{x}')] \quad (4)$$

Plain gradients fail this axiom due to saturation at nonlinear activations such as ReLU, where zero gradients prevent full reconstruction of the output difference. Expected gradients satisfies completeness via the gradient theorem (the fundamental theorem of calculus for line integrals), with the integrated quantity being the expectation of the model output over the background distribution.

- **Linearity:** For a model that is a linear combination of two submodels, $f(\mathbf{x}) = af_1(\mathbf{x}) + bf_2(\mathbf{x})$, the attributions decompose accordingly as $\phi(\mathbf{x}) = a\phi_1(\mathbf{x}) + b\phi_2(\mathbf{x})$. This follows directly from the linearity of gradients.
- **Symmetry-preserving:** If two features are symmetric, meaning they enter the model identically and take identical values in the input then they receive identical attributions. Expected gradients inherits this property because it is a symmetric function of symmetric functions.

These axioms collectively distinguish expected gradients from simpler attribution approaches such as plain gradients or gradient $\times$ input, and provide formal guarantees that are particularly valuable when attributions are used to draw biological conclusions. For instance, when ranking genes by their contribution to predicted cell-type proportions in a deconvolution setting.

B. Dataset and Model Selection

We applied SHAP-based interpretability analysis to DISSECT [11] models from a previously published

benchmarking study [2]. That study evaluated multiple reference-target matching configurations; here we use the healthy-reference case, where scRNA-seq from healthy donors served as the deconvolution reference. Model performance was assessed on 80 pseudo-bulk test samples spanning nine immune cell types: B cells (B), CD4 T cells (T4), CD8 T cells (T8), natural killer cells (NK), proliferating cells (Prolif), classical monocytes (cM), non-classical monocytes (ncM), plasmacytoid dendritic cells (pDC), and conventional dendritic cells (cDC). Of these, 40 were generated from healthy held-out single-cell data and 40 from SLE patient-derived single-cell data. We selected the best-performing simulation split based on mean Lin’s concordance correlation coefficient (CCC) and median root mean squared error (RMSE) across the major cell types.

In the benchmark study [2], DISSECT was trained as an ensemble of five networks (four hidden layers: 512→256→128→64, ReLU6 activations, softmax output) on 9,000 simulated bulk samples (450 per donor from 20 healthy reference donors), where each sample was drawn from a single donor to avoid introducing batch effects during training. Training followed a two-phase schedule: an initial regression phase minimizing KL divergence between predicted and simulated proportions, followed by a consistency phase that enforced linearity of predictions under convex combinations of simulated and real bulk samples. The consistency anchor was a single healthy pseudo-bulk sample. All subsequent SHAP analyses use this ensemble of five models, applied to 80 pseudo-bulk test samples (40 healthy, 40 SLE) profiled across 20,473 genes. Because the SLE reference dataset contained a reduced gene set (19,467 genes) relative to the healthy reference (20,473 genes), SLE pseudo-bulk samples used in run 2c were projected into the healthy reference gene space prior to normalization, with missing genes zero-filled.

C. Experimental Design

The key principle underlying our design is that SHAP attributions are always defined relative to a background distribution: a gene receives high attribution when its expression in the target samples causes the model’s prediction to deviate substantially from what the background suggests. By systematically varying the background-target pair across eight experimental conditions (Table I), we decompose gene attributions into three biologically interpretable components. Figure 1 provides a conceptual overview of how different background choices isolate raw marker signal, domain shift, and disease-associated condition shift.

- **Raw cell-type marker signal (RCM):** Using an uninformative background such as random noise (runs 1a, 2a) or all-zeros (runs 1c, 2e), attributions reflect genes whose absolute expression levels drive predictions beyond a non-biological baseline. Since neither background carries cell-type-specific information, high-attribution genes are expected to represent

TABLE I: SHAP experimental conditions. All runs use normal-reference models. RCM = raw cell-type marker signal; DS = domain shift (pseudo-bulk $\rightarrow$ real bulk); CS = condition shift (healthy $\rightarrow$ SLE).

Run	Patients	Background	Captures
1a	Healthy	Noise	RCM (noise)
1b	Healthy	Sim. healthy	DS
1c	Healthy	Zero	RCM (zero)
2a	SLE	Noise	RCM (noise)
2b	SLE	Sim. healthy	DS + CS
2c	SLE	Sim. SLE	DS
2d	SLE	Real healthy	CS
2e	SLE	Zero	RCM (zero)

canonical markers the model has learned for each cell type. Comparing runs 1a and 1c allows us to assess whether the choice of uninformative background affects marker recovery.

- Domain shift (DS): Using simulated pseudo-bulk as the background (runs 1b, 2c), the baseline reflects what the model has already learned from its training distribution. Attributions then capture how each gene’s real-tissue expression deviates from pseudo-bulk expectations, isolating the systematic gap between simulated training data and real tissue. Comparing runs 1b and 2c reveals whether this domain shift is consistent across healthy and SLE conditions.
- Condition shift (CS): Using real healthy pseudo-bulk as the background for SLE patients (run 2d), attributions isolate genes whose expression in SLE deviates from the healthy baseline, capturing disease-specific signals. Run 2b captures both DS and CS simultaneously; comparing 2b and 2c therefore allows the condition-specific component to be further decomposed.

D. SHAP-Based Interpretation of DISSECT Predictions

SHAP values were computed using the Gradient-Explainer implementation from the SHAP Python library [12], which approximates Shapley values via expected gradients [15]. All samples were normalized using DISSECT’s `normalize_per_batch` function (log1p followed by per-sample min-max scaling) before SHAP computation to match training-time preprocessing. Values were computed independently for each of the five ensemble models and averaged element-wise; global feature importance for each cell type was then summarized as the mean absolute SHAP value across all test samples.

SHAP beeswarm plots and mean absolute SHAP bar plots were generated for the top-ranked genes within each cell type.

To assess biological plausibility of the learned feature attributions, we curated 26 canonical marker genes spanning all nine immune cell types based on established literature (Supplementary Table 1). For each marker gene and each

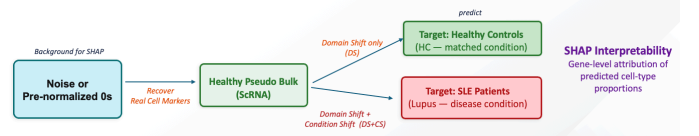


Fig. 1: Conceptual overview of the SHAP experimental design used to interpret DISSECT predictions. Different SHAP background distributions were used to investigate how gene attributions change under varying biological contexts. Noise or pre-normalised backgrounds should emphasize global marker recovery, while simulated healthy pseudo-bulk establishes a biologically realistic baseline. Comparisons between healthy and lupus bulk samples against healthy pseudo-bulk backgrounds capture domain shift (DS) and disease-associated condition shift (DS+CS), respectively. SHAP values quantify gene-level contributions to predicted cell-type proportions

TABLE II: DISSECT-predicted mean cell-type proportions in healthy and SLE pseudo-bulk samples.

Cell Type	Healthy	Lupus	Δ
T4	0.361	0.231	-0.130
cM	0.179	0.304	+0.125
Prolif	0.025	0.037	+0.011
B	0.110	0.087	-0.023
ncM	0.055	0.071	+0.016
T8, NK, cDC, pDC	—	—	<0.01

experimental condition, we recorded the rank of its mean absolute SHAP value among all 20,473 genes, where lower rank indicates greater contribution to the predicted cell-type proportion.

IV. Experimental Results

With eight SHAP runs and nine immune cell types, a fully comprehensive presentation of results would span 72 condition–cell-type combinations, each with thousands of ranked genes. For clarity, we focus on a single representative cell type to trace all three biological components (RCM, DS, and CS) through the same lens.

To motivate the choice, we summarize predicted cell-type proportions from the DISSECT model across 40 healthy and 40 SLE patients (Table II). T4 are most strongly depleted in SLE ($\Delta = -0.130$) and cM are most strongly expanded ($\Delta = +0.125$), findings consistent with established lupus immunology [2]. Beyond proportion shifts, the same dataset shows condition-associated expression changes within B, T8, and ncM [2]. Rather than fixing a single focal cell type across all three components, we select the most illustrative cell type for each: T4 and cM for RCM, where canonical marker recovery is clearest; T4 for DS, where the domain gap is most pronounced; and cM for CS, where the SLE-associated inflammatory signal is strongest.

TABLE III: Canonical marker ranks (out of 20,473 genes) under noise (run 1a) and zeros (run 1c) backgrounds. Negative rank indicates a negative mean SHAP value for that gene.

Cell Type	Marker	1a Rank	1c Rank
T4	CD4	-10,411	199
T4	CD40LG	-324	4
T4	IL7R	14,433	46
Prolif	MKI67	-1	2
Prolif	TK1	-8	6
Prolif	BIRC5	-18	21
cM	CD14	16,752	6
cM	MCEMP1	-46	16
cM	CLEC4E	-114	2

A. RCM: Recovering Canonical Cell-Type Markers

Table III and Figure 2 compare canonical marker ranks and per-sample SHAP values between the noise (run 1a) and zeros (run 1c) backgrounds for three representative cell types: T4, Prolif, and cM. The zeros background recovers canonical markers substantially more reliably. For most markers, the noise background assigns either low positive ranks or negative SHAP values (indicating the gene suppresses the prediction under that baseline), whereas the zeros background consistently places the same genes near the top. Prolif markers (MKI67, TK1, BIRC5) are the exception: they rank near the top under both backgrounds, reflecting their uniquely high and cell-type-specific expression. Across all nine cell types, GO enrichment of the top-attributed genes increased from 372 significant terms under noise to 3,276 under zeros, confirming that zero is the more biologically informative uninformative baseline (Supplementary Table 2). Supplementary Figure 3 illustrates this contrast for B, where the zeros background recovers canonical B cell activation and proliferation terms that are entirely absent under noise.

The reason zero outperforms noise is that a noise background introduces random positive expression as the reference level: any gene whose real expression falls below the noise baseline receives a negative attribution, regardless of its biological relevance, while non-specific high-expression genes are falsely credited simply for exceeding a high random baseline. A zero background, by contrast, sets the reference at complete silence, so only genes with active positive expression in the target samples receive positive attribution, which naturally selects for the expressed canonical markers that define each cell type.

B. DS: Genes Driving the Domain Shift

With a simulated pseudo-bulk background (runs 1b and 2c), attributions capture genes whose expression in real tissue deviates from what was seen during training, isolating the domain shift between simulated and real pseudo-bulk. Figure 3 shows the top 20 attributed genes for T4 under run 1b. Two types of genes emerge simultaneously: canonical markers and non-typical genes. CD40LG, a well-established T4 marker, ranks 2nd, confirming that

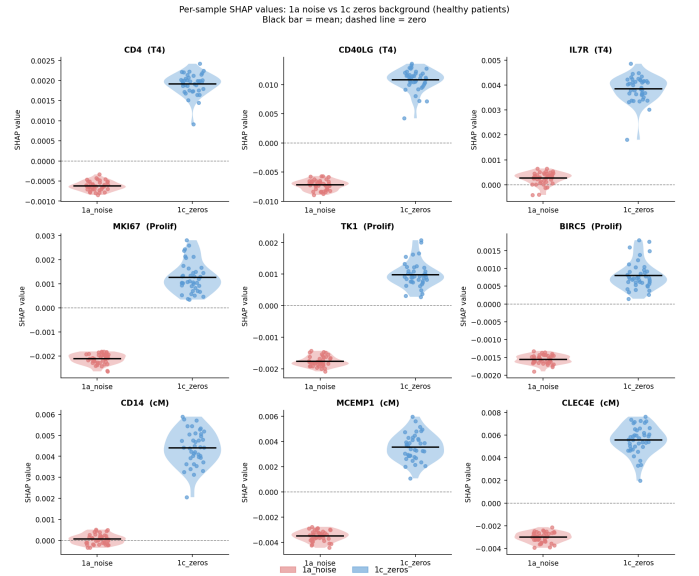


Fig. 2: Per-sample SHAP values for canonical markers under noise (run 1a, pink) and zeros (run 1c, blue) backgrounds. Each point is one healthy patient sample; the black bar shows the mean. The dashed line marks zero. Zeros background yields consistently positive, higher-magnitude attributions for all markers except Prolif, which ranks highly under both backgrounds.

the simulated background still recovers biologically meaningful signal. However, the top-ranked gene is TTC39C-AS1, a long non-coding RNA with no established T4 identity role, and most remaining top genes (ADTPR, ANKRD55, KRT1, ST6GALNAC1) are similarly non-canonical. These genes likely reflect systematic expression differences between simulated pseudo-bulk and real tissue that the model has learned to exploit, rather than genuine cell-type markers. This pattern reflects the DISSECT training objective: the network discriminates between cell types, so SHAP credits genes that resolve ambiguity between similar populations as well as genes that differ between the simulation and reality.

To assess how much of the domain shift is condition-specific, we compared the top 500 SHAP genes between runs 1b (healthy patients) and 2c (SLE patients), both using a matched simulated background (Figure 4). Approximately 55% of genes overlapped across all nine cell types, indicating that more than half of the domain shift signal is shared between healthy and SLE conditions and is therefore a consistent property of the simulation-to-real gap. This convergence is also reflected at the pathway level: the top 15 GO biological process terms for T4 are largely consistent between runs 1b and 2c, with both dominated by T cell activation, lymphocyte differentiation, and leukocyte-mediated immunity terms (Supplementary Figure 4). The remaining 45% is condition-specific and is further characterized by the CS analysis below.

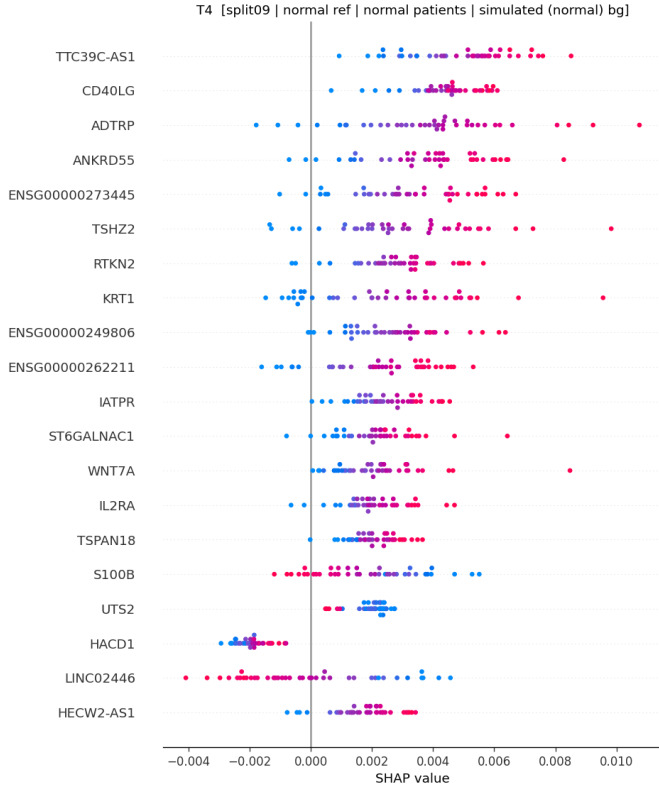


Fig. 3: Top 20 SHAP-attributed genes for T4 under run 1b (healthy patients, simulated healthy pseudo-bulk background). Each dot is one patient sample; color indicates the gene’s expression level (pink = high, blue = low). CD40LG, a canonical T4 marker, ranks 2nd, while the majority of top genes are non-canonical, reflecting domain shift artifacts.

C. CS: Genes Mediating the Condition Shift

Run 2d uses real healthy pseudo-bulk as the background for SLE patients, so high-attribution genes are those whose expression in SLE deviates specifically from a healthy baseline, capturing the condition shift. Figure 5 shows the top 20 attributed genes for cM under run 2d, and the genes cluster into four biologically coherent groups that reflect the known pathological role of monocytes in SLE [16].

Immune complex clearance. FCGR1A (CD64), the top-ranked gene, encodes the high-affinity IgG Fc receptor on monocytes that mediates phagocytosis of immune complexes, a central driver of monocyte activation in SLE. FCAR (CD89), an IgA Fc receptor, further supports this role.

Type I interferon imprinting. SIGLEC1 (CD169) is among the most strongly interferon-stimulated genes on monocytes and is a well-established biomarker of type I IFN activity in SLE [17]. Its high attribution in run 2d reflects the IFN-driven transcriptional state of SLE monocytes that is absent in the healthy background.

Damage sensing. CLEC4E (Mincle) and CLEC4D are C-type lectin receptors activated by damage-associated

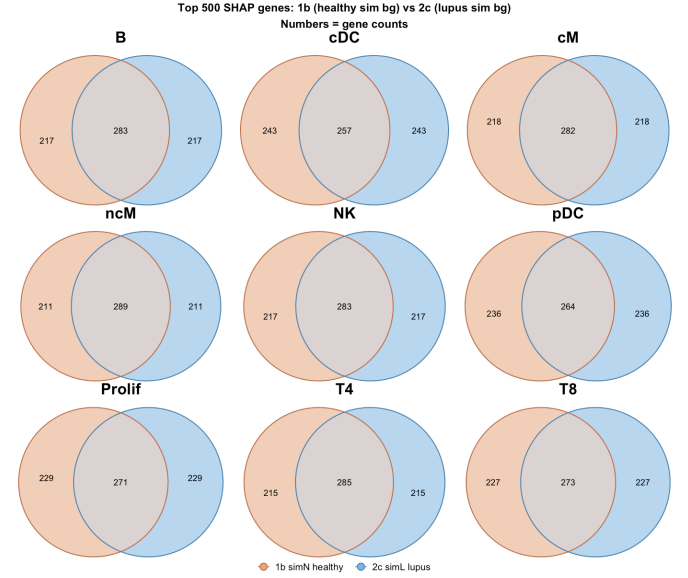


Fig. 4: Overlap of the top 500 SHAP genes between run 1b (healthy patients, simulated healthy background) and run 2c (SLE patients, simulated SLE background) for each of the nine cell types. Each circle represents one run; the overlapping region shows genes shared between the two domain-shift conditions.

molecular patterns (DAMPs) released from necrotic or apoptotic cells, connecting impaired apoptotic clearance to downstream monocyte activation.

Inflammatory amplification. IL1B encodes interleukin-1 β , a master pro-inflammatory cytokine produced by activated monocytes, and CXCL2 is a neutrophil-recruiting chemokine that propagates the inflammatory cascade. THBS1 (thrombospondin-1) contributes to both apoptotic cell clearance and TGF- β activation, linking tissue damage to immune amplification.

Run 2d produced the richest GO enrichment across all conditions, with 3,226 significantly enriched biological process terms in total, consistent with the broad and coordinated transcriptional reprogramming of cM in SLE. The top 15 GO biological process terms for cM (Supplementary Figure 5) reinforce these gene-level findings: chemotaxis, leukocyte migration, phagocytosis, myeloid leukocyte activation, and positive regulation of inflammatory response are among the most significantly enriched, directly corresponding to the CXCL2, FCGR1A, CLEC4E, and IL1B signals observed in the beeswarm plot.

V. Discussion

This work demonstrates that SHAP, applied with a carefully chosen background distribution, can recover biologically meaningful signals from a black-box deconvolution model. The three-part experimental design reveals complementary aspects of model behavior: uninformative backgrounds expose canonical marker recovery, simulated backgrounds isolate the domain gap between training and inference, and a real healthy background captures

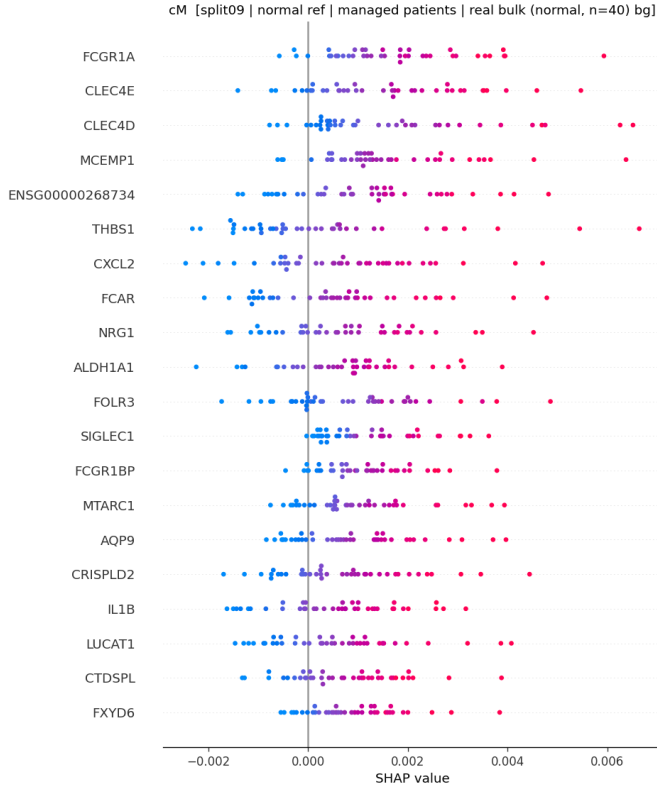


Fig. 5: Top 20 SHAP-attributed genes for cM under run 2d (SLE patients, real healthy pseudo-bulk background). Each dot is one SLE patient sample; color indicates expression level (pink = high, blue = low). Genes group into four functional clusters: immune complex clearance (FCGR1A, FCAR), type I interferon imprinting (SIGLEC1), damage sensing (CLEC4E, CLEC4D), and inflammatory amplification (IL1B, CXCL2, THBS1).

disease-specific transcriptional changes in SLE. Together, these findings suggest that DISSECT has internalized genuine cell-type biology rather than relying on spurious correlates, which strengthens confidence in applying its predictions to immunological research.

A recurring observation across all conditions is that the model favors discriminative over identity markers. For T cell subtypes, classical pan-T markers such as CD3D ranked poorly, while genes that differentiate CD4 from CD8 cells, such as CD40LG, ranked highly. This reflects the DISSECT training objective, which optimizes for proportion prediction across competing cell types rather than absolute expression of canonical genes. The implication for users is that SHAP attributions from deconvolution models should be interpreted as discriminative importance scores, not expression signatures, and biological interpretation should account for the multi-class structure of the prediction task. Several limitations should be noted. First, all analyses were performed on pseudo-bulk samples constructed by aggregating held-out single cells, rather than on independently collected real bulk RNA-seq data. Pseudo-bulk samples share the same

single-cell reference as the training data, which means the domain gap between training and test is smaller than what would be encountered in practice. As a result, the domain shift signals captured here may underestimate the complexity of real-world deconvolution scenarios. Applying the same SHAP pipeline to matched real bulk and pseudo-bulk samples from the same donors would allow a direct comparison and provide a more stringent test of what the model has generalised.

Second, gene collinearity poses a challenge for interpreting individual SHAP attributions. SHAP values assume that features can be marginalised independently, but co-expressed genes violate this assumption: when two genes are highly correlated, Shapley-based methods may concentrate credit on one arbitrarily while under-attributing the other, even if both are equally relevant. In immune transcriptomics, where many canonical markers belong to tightly co-regulated modules (e.g., interferon-stimulated genes, Fc receptor family members), this means individual gene ranks should be interpreted with caution and pathway-level enrichment provides a more robust signal.

Third, SHAP values are computed separately for each cell-type output, so a gene that appears important for one cell type may incidentally co-vary with others due to shared transcriptional programmes. The current analysis treats each cell type independently; a richer interpretation would require examining cross-cell-type attribution patterns and the internal representations of individual ensemble members to understand how the model resolves ambiguity between similar populations such as cM and ncM, or T4 and T8.

VI. Conclusion

We applied SHAP-based interpretability analysis to DISSECT, a deep learning cell-type deconvolution model, using a systematic background-variation strategy to decompose gene attributions into three biologically distinct components. Using uninformative backgrounds (noise and zeros), we showed that DISSECT recovers canonical immune cell-type markers, with zero backgrounds providing substantially cleaner signal than noise. Using simulated pseudo-bulk backgrounds, we identified a consistent domain shift shared across healthy and SLE conditions, accounting for roughly 55% of the top attributed genes. Using real healthy pseudo-bulk as the background for SLE patients, we isolated condition-specific transcriptional changes including monocyte inflammatory pathways, NK cell chemokine signaling, and pDC-driven interferon responses.

These results establish that SHAP with appropriate background selection is a practical and biologically informative tool for interpreting neural network deconvolution models. The framework is general and could be applied to other DISSECT configurations or extended to real bulk RNA-seq data, where the domain shift and condition shift signals are expected to be larger and potentially more informative.

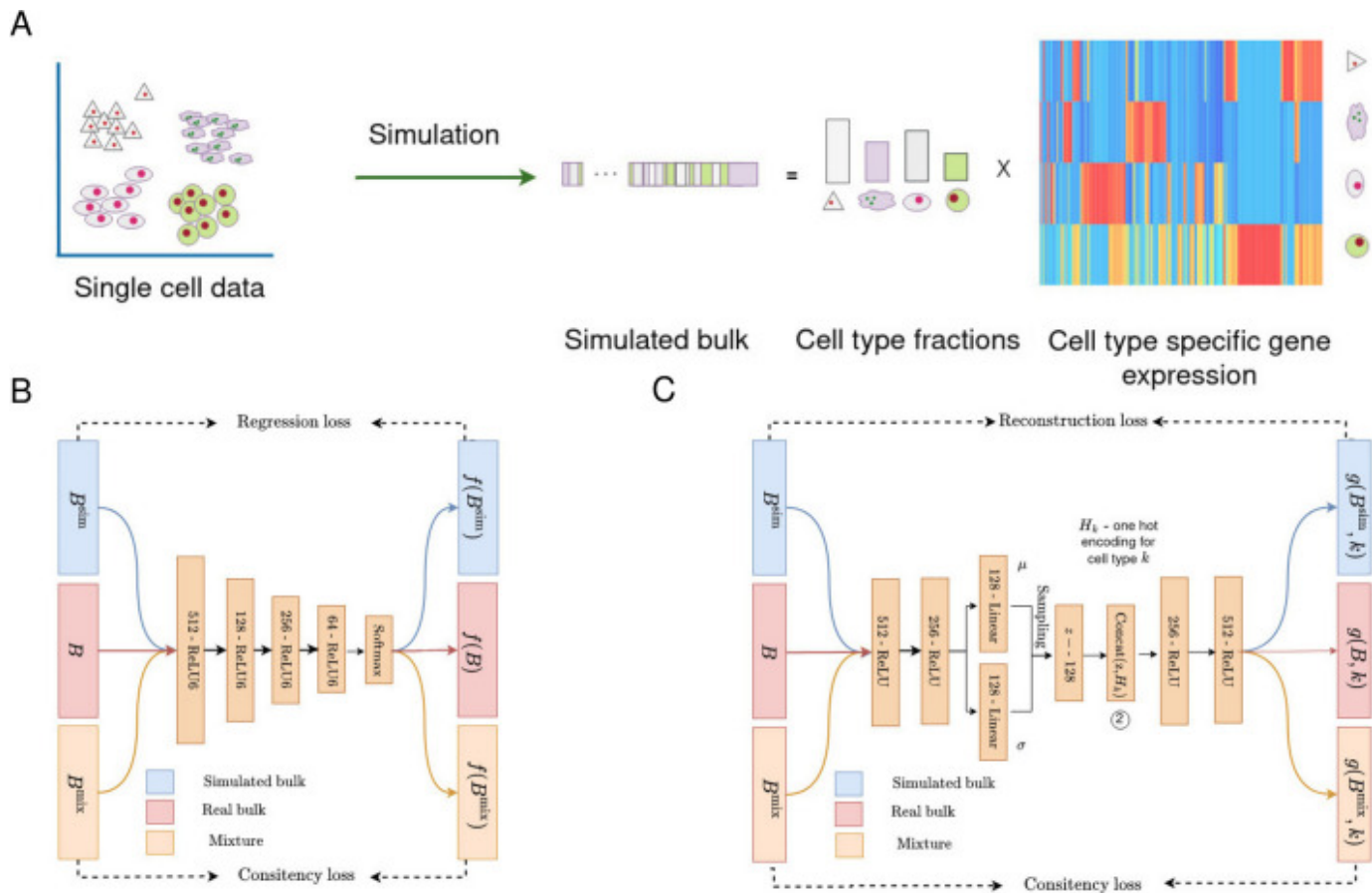
VII. Author Contributions

Zhao Y.: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft (Results, Discussion). Shah J.: Investigation, Writing - Original Draft (Introduction, Methods), Writing - Review & Editing, Visualization. All authors read and approved the final manuscript.

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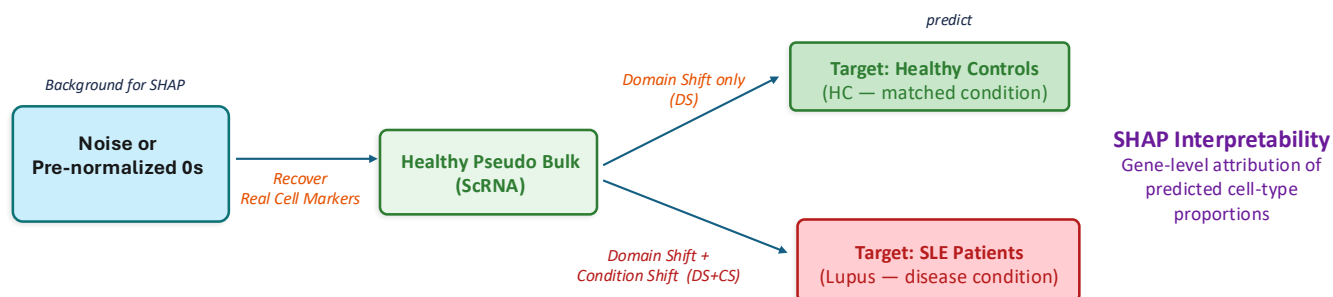
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Appendix

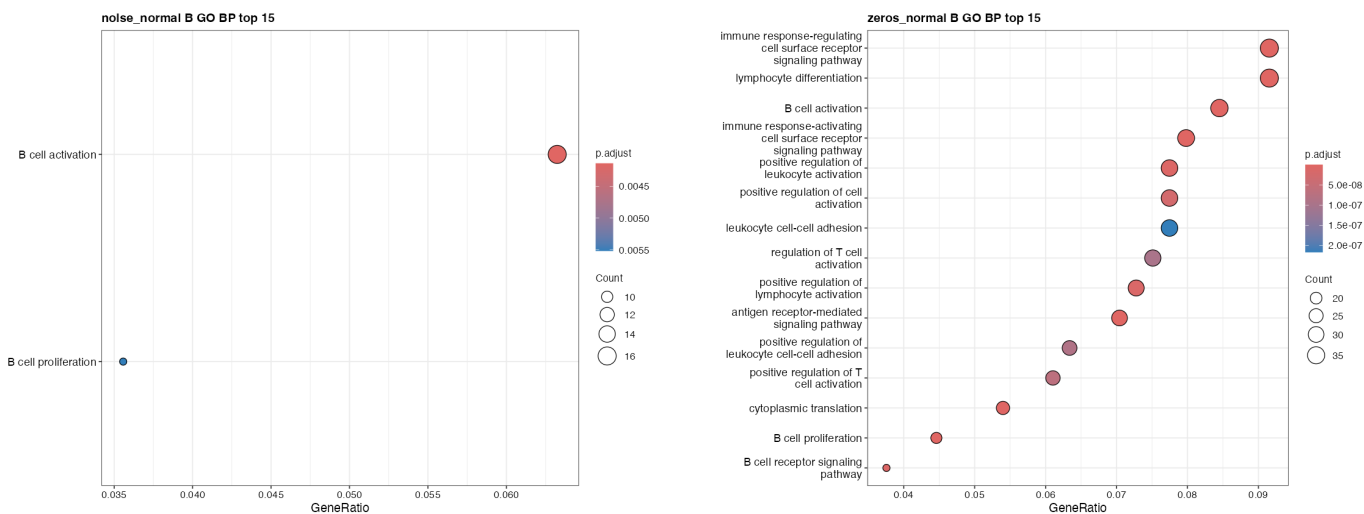


Supplementary Figure 1: Overview of the DISSECT framework

A Illustration of the simulation procedure using reference single-cell data. The figure shows the simulation of one sample which consists of cell type fractions, simulated gene expression and cell type specific gene expression profiles (i.e., signature matrix). B Detailed overview of an MLP used to estimate cell type fractions. C Overview of an autoencoder used to estimate cell type specific gene expression profiles



Supplementary Figure 2: Overview of the SHAP experimental design.



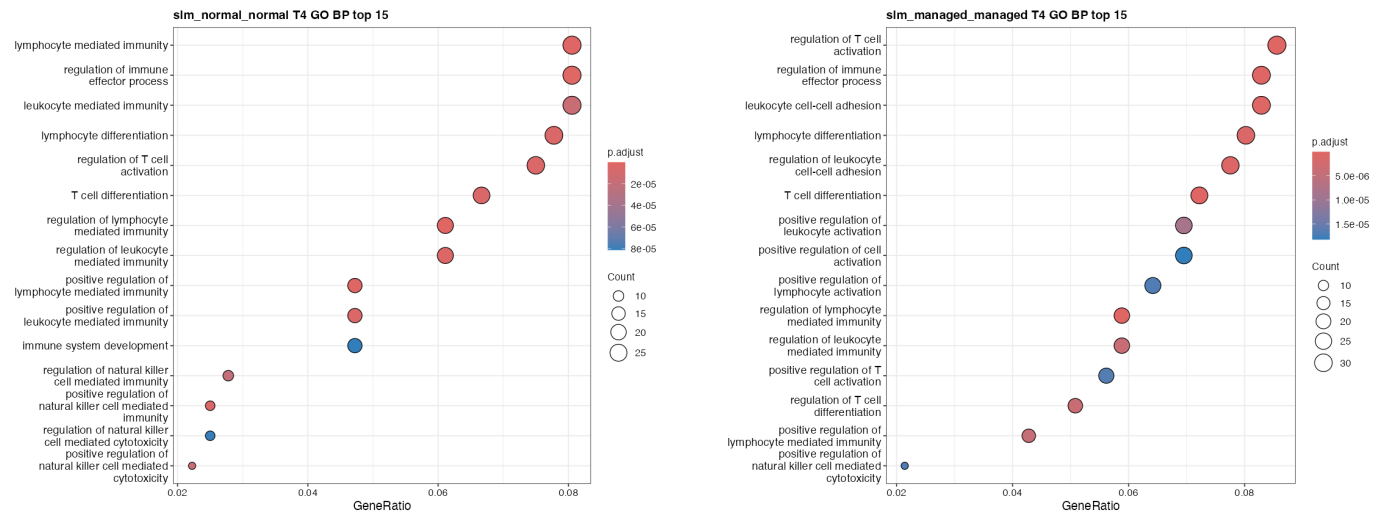
Supplementary Figure 3: GO biological process enrichment for B under noise (run 1a, left; 2 significant terms) and zeros (run 1c, right; 160 significant terms) backgrounds. Canonical B cell terms (B cell activation, B cell proliferation) are highlighted by red boxes in the zeros panel.

Supplementary Table 1: Canonical markers used to evaluate SHAP attribution quality across nine cell types.

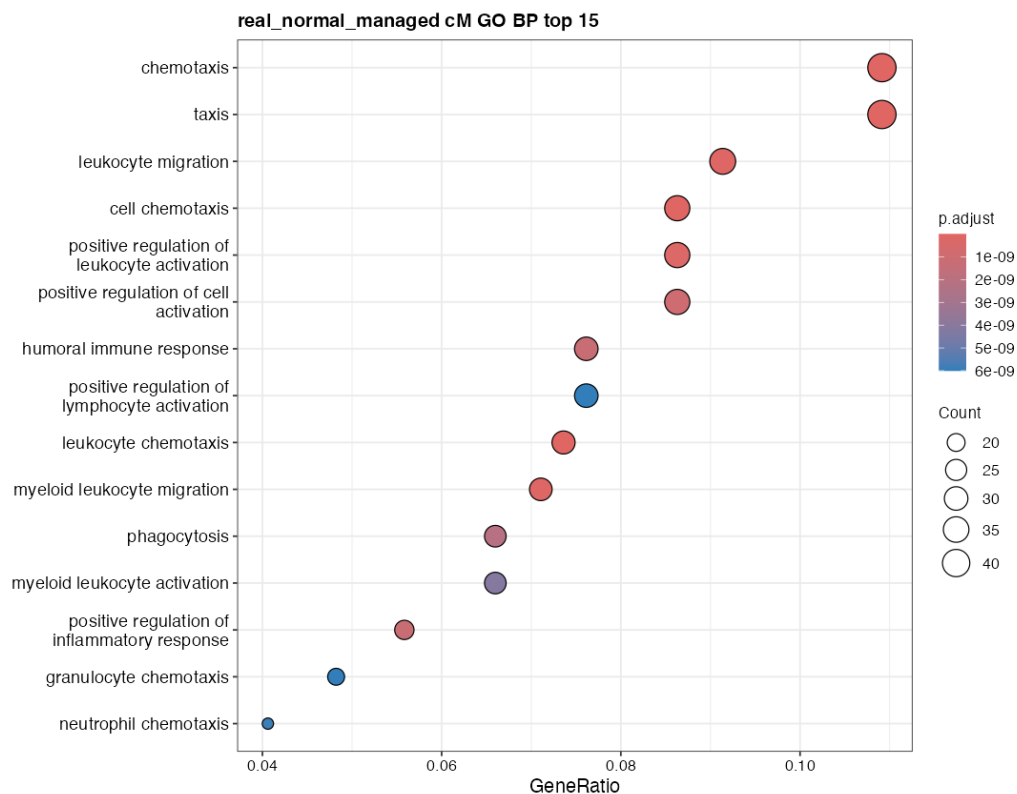
Cell Type	Gene	Ensembl ID	Role
B cell	CD19	ENSG00000177455	B cell identity; pan-B surface receptor
	MS4A1	ENSG00000156738	B cell identity (CD20); surface antigen
	FCER2	ENSG00000104921	B cell activation marker (CD23)
CD4 T	CD4	ENSG0000010610	CD4 T cell co-receptor
	CD40LG	ENSG00000102245	T helper activation (CD154); CD4-enriched
	IL7R	ENSG00000168685	T cell survival receptor (CD127)
CD8 T	CD8A	ENSG00000153563	CD8 T cell co-receptor (alpha chain)
	CD8B	ENSG00000172116	CD8 T cell co-receptor (beta chain)
	LAG3	ENSG00000089692	Immune checkpoint; exhausted CD8 T cells
NK	NCAM1	ENSG00000149294	NK identity (CD56); canonical NK marker
	NCR1	ENSG00000189430	NK activating receptor (NKp46)
	KIR3DL2	ENSG00000240403	Inhibitory KIR; NK dysfunction in SLE
Prolif	MKI67	ENSG00000148773	Proliferation marker (Ki-67)
	TK1	ENSG00000167900	DNA synthesis; S-phase marker
	BIRC5	ENSG00000089685	Survivin; cell cycle / anti-apoptosis
cM	CD14	ENSG00000170458	Classical monocyte identity
	MCEMP1	ENSG00000183019	Monocyte-enriched surface protein
	CLEC4E	ENSG00000166523	Mincle; classical monocyte activation
ncM	CDKN1C	ENSG00000129757	Non-classical monocyte identity (p57)
	CX3CR1	ENSG00000168329	Fractalkine receptor; ncM patrolling
pDC	IL3RA	ENSG00000185291	pDC identity (CD123)
	CLEC4C	ENSG00000198053	pDC identity (BDCA-2)
	LILRA4	ENSG00000239961	pDC activation; upregulated in SLE
cDC	CLEC9A	ENSG00000197992	cDC1 identity (DNDR-1)
	CLEC10A	ENSG00000132514	cDC2 identity
	FCER1A	ENSG00000179639	High-affinity IgE receptor; cDC2/mast cells

Supplementary Table 2: Number of significantly enriched GO biological process terms for each cell type under noise (run 1a) and zeros (run 1c) backgrounds, and the improvement.

Cell Type	1a noise bg	1c zeros bg	Improvement
ncM	4	518	514
cM	59	522	463
T4	12	386	374
cDC	6	349	343
pDC	11	335	324
NK	33	312	279
T8	19	265	246
Prolif	226	429	203
B	2	160	158



Supplementary Figure 4: Top 15 GO biological process terms for T4 under run 1b (healthy patients, simulated healthy background, left) and run 2c (SLE patients, simulated SLE background, right). Both panels are dominated by T cell activation, lymphocyte differentiation, and leukocyte-mediated immunity terms, consistent with the ~55% gene-level overlap between the two runs and confirming that the domain shift signal is largely condition-independent.



Supplementary Figure 5: Top 15 GO biological process terms for cM under run 2d (SLE patients, real healthy pseudo-bulk background), capturing the condition shift. Chemotaxis, leukocyte migration, phagocytosis, myeloid leukocyte activation, and positive regulation of inflammatory response are among the most significantly enriched terms (adjusted $p < 10^{-9}$), consistent with the CXCL2, FCGR1A, CLEC4E, and IL1B signals identified in the beeswarm plot.